



NATURAL FIBER COMPOSITE · AHIMSA DESIGN

Only when we make the right
choices **do we achieve the right
results.**

TERMITE PROOF

WATER PROOF

FLAME RETARDANT

EASILY MACHINABLE

THERMOFORMABLE

ECO-FRIENDLY & RECYCLABLE

THE REAL DESIGN MANTRA

We don't **destroy to design.**
We don't **kill to create.**
We don't **cut down to build.**

If you love something, set it free — and it will still
work. That's **Ahimsa**.

FROM THE CHAIRMAN

"Embracing a sustainable future, one nfc board at a time."



A handwritten signature in black ink, appearing to read 'B. L. Bengani'.

B. L. Bengani

CHAIRMAN, INDOWUD NFC

With over 30 years in the ply industry, Mr. Bengani — founder of Uniply — is pioneering natural-fibre composite wood in India and the world.

In a world grappling with the escalating climate crisis, every responsible citizen and business leader must actively seek solutions. At Indowud nfc, we believe innovation and sustainability are not buzzwords, but the cornerstones of building a better tomorrow.

It is with immense honour that I present our remarkable range of sustainable products. Crafted from a naturally abundant and renewable resource, they are eco-friendly with superior qualities — the ideal choice for home owners, architects, designers, builders and contractors.

Our meticulous manufacturing ensures products that are termite-proof, water-proof, and even flame-retardant + smoke-suppressant — suitable for a wide range of interior and exterior applications.

By choosing our nfc products, you join us on this journey — embracing the future, embracing sustainability. Together, let's build a world where beauty and responsibility go hand in hand.

A PIONEERING WOOD

That lets the nation breathe freely.

Indowud nfc is the first-of-its-kind agricultural-husk wood. This husk is often thrown away or burnt — causing alarming air pollution across India and the world.

Reusing it is of great value to national health and breathable air. With state-of-the-art technology these husks become boards that mimic every value of wood — unmatched in India and, probably, the world.

0 TREES CUT,
EVER

1st AGRICULTURAL-
HUSK WOOD OF
ITS KIND

100% OF THE WOOD
VALUES, NONE
OF THE FELLING

THE PHILOSOPHY

Ahimsa Design Philosophy

No trees are cut when making Indowud nfc.
India has one of the worst termite problems
— but nfc gives them nothing to eat. You don't
even need to kill a single termite.

A**Axe the axe**

No more trees cut — not one.

H**Healthy homes**

Anti-bacterial protective properties.

I**Ice, rain & water proof**

Zero swelling. Zero delamination.

M**Mouldable**

Thermoformable and easy to design with.

S**Secure**

Against termites and rodents alike.

A**Agricultural-husk made**

Preventing air pollution from husk burning.

BUILDING SUSTAINABLE & HEALTHY HOMES

A protective shield, engineered in.

Non-hazardous additives create an anti-bacterial shield during manufacturing. Indowud nfc is non-toxic, emits no VOCs, and is engineered to keep bacteria away.

Our home is the only sanctuary we retreat to — let us make it safe, secure and healthy.

✓ NON-TOXIC☞ ZERO VOC✓ ANTI-BACTERIAL✓ LEAD-FREE

GreenPro Ecolabel

CII has certified Indowud nfc as a Sustainable Green Product and awarded the GreenPro ecolabel — accredited by the Global Ecolabelling Network (GEN) through GENICES.

Products bearing GreenPro have lower environmental impact and enhance the performance of Green Buildings and Green Companies.

ISO 9001:2015ISO 14001:2015ISO 45001:2018GREENPROLEAD-FREE

EIGHTEEN REASONS, ONE MATERIAL

Everything plywood promises — without the compromises.

A single engineered material that answers for structure, safety, health and the planet at once.



Termite proof



Water proof



Flame retardant



Easily machinable



No splintering or cracking



Durable



Anti-rodent



Good screw holding



Fungus, algae & mould resistant



Anti-bacterial



UV resistant



Thermoformable



Smoke suppressant



100% recyclable



No harmful ingredients



Absorbs sound



100% eco-friendly



No formaldehyde emission

THE RANGE

Eleven products. One zero-wood system.

Eco-friendly products for a healthier planet — across every surface, interior and exterior. A page for each follows.

**zerOwud nfc board**

PANEL

**Indowud nfc board**

HERO PANEL

**nfc door**

JOINERY

**nfc frame**

JOINERY

**nfc decking**

OUTDOOR

**nfc flute**

LOUVER

01 · ZEROWUD NFC

zerOwud nfc board

Build green with zero-wud panels

ZerOwud boasts good strength & durability — ideal for everything from furniture and cabinetry to wall panels. For more, visit zerOwud.com.

- ✓ STRONG
- 🛡️ DURABLE

STANDARD SIZE	8ft × 4ft · 2440 × 1220 mm
THICKNESS	6 · 8 · 12 · 16 · 18 · 25 mm
SURFACE	Smooth, both sides

Contact us for custom sizes.



ZEROWUD · SMOOTH BOTH SIDES

02 · INDOWUD NFC

Indowud nfc board

Transforming agricultural waste into a sustainable panel

A sustainable, high-performing alternative to traditional wood. Durable and versatile — ideal across interiors and exteriors.

-  NEO
-  CREATE
-  BUILD

RANGE	nfc Neo · Create · Build
AVAILABLE SIZE	8ft × 4ft · 2440 × 1220 mm
THICKNESS	6 · 8 · 12 · 15/16 · 18 · 25 mm
SURFACE	Rough, both sides

Contact us for custom sizes.



Created for design. Mouldable, yet sturdy.

A superior alternative to wood in many ways — it looks, feels and works like wood too.



Office furniture



Cabinets & wardrobes



Moist areas & bathroom



Roofs



Stairs



Kitchen & dining

APPLICATIONS & INSPIRATION · DESIGNED BY TOP ARCHITECTS

Converting imagination to reality.

From sculpted swings to seamless interiors — Indowud nfc bends to the architect's vision without bending the rules of sustainability.

SWINGS

INTERIORS

BESPOKE JOINERY

CURVED FORMS

APPLICATIONS & INSPIRATION · EXTERIORS

Built for extreme weather. Far superior to any wood-based panel.

Use it freely across all exteriors and interiors.



Pool decks



Benches



Swings



Doors



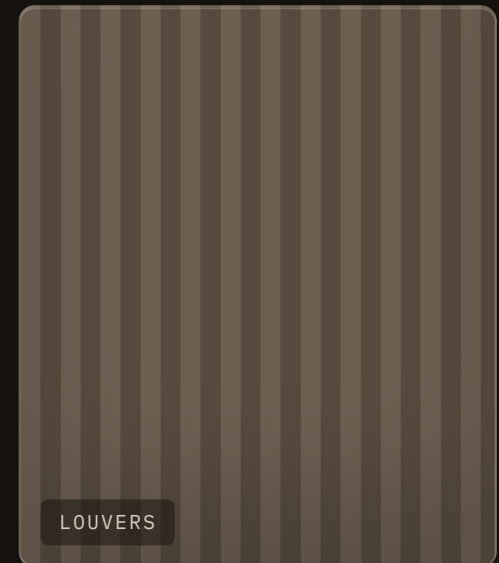
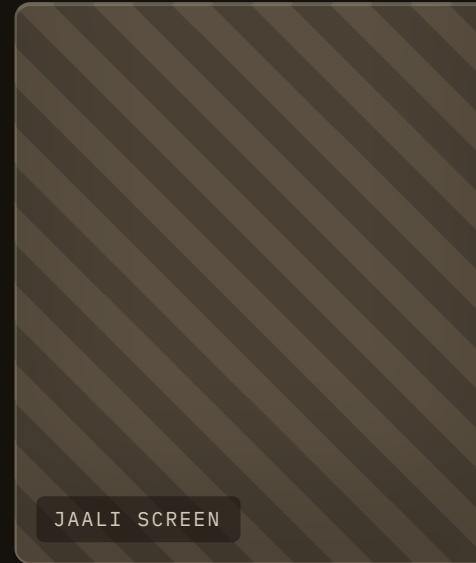
Fencing



Outdoor furniture

APPLICATIONS & INSPIRATION · REAL PROJECTS

Building beautiful exteriors.



03 · NFC DOOR

nfc door

The eco-friendly door

Combine the timeless beauty of a natural wood panel with sustainability. Easily paint, polish, varnish, or overlay veneer / laminate.

SIZE	7ft × 3ft · 2140 × 920 mm
ALSO	8ft × 3ft · 2440 × 920 mm
THICKNESS	28 mm & 30 mm



04 · NFC FRAME

nfc frame

Reimagine doorways with nfc frames

Door frames with open grains akin to natural wood.
Customise with ease — easy to paint, varnish and stain —
effortlessly merging beauty and sustainability.

SIZES **3"×2" · 4"×2.5" · 5"×2.5"**

STANDARD LENGTH **7ft · 8ft · 10ft**

FINISH **Paint · varnish · stain**

Download the nfc-frame brochure at indowud.com/downloads/



NFC FRAME · OPEN GRAIN

05 · NFC JAALI

nfc Jaali

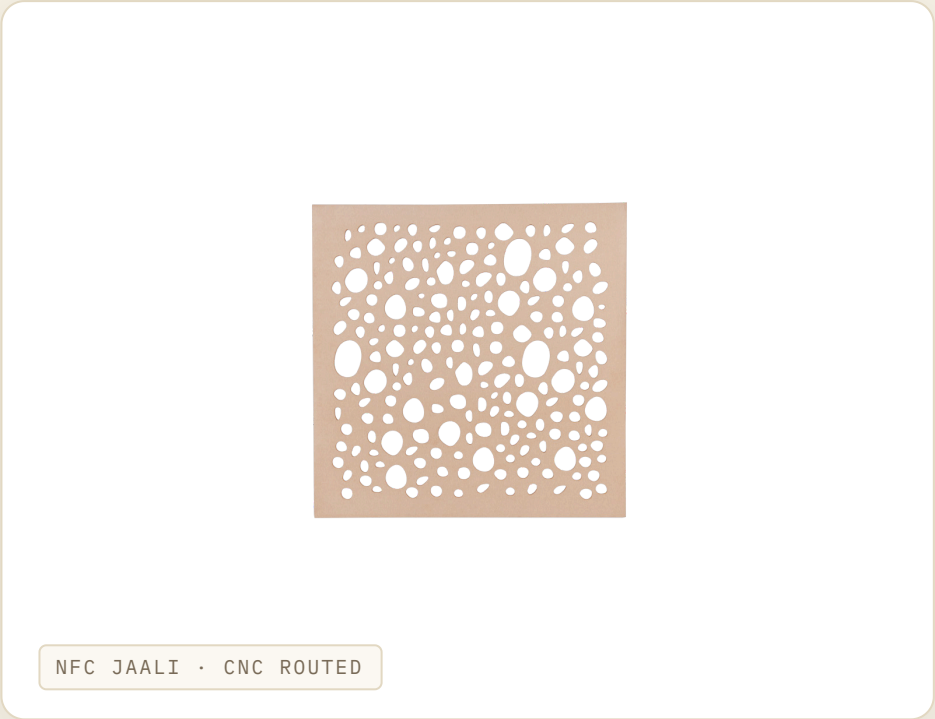
Elevate your space with intricate patterns

A fusion of art, sustainability and strength. CNC-routed nfc boards — including stunning mashrabiya screens — enhance interiors and lighten the weight of building exteriors.

- CNC ROUTED
- MASHRABIYA

METHOD	CNC routing
USE	Interior screens · exterior facades
DESIGNS	Fully customisable

Contact us for customisation of designs.



06 · NFC DECKING

nfc decking

Elevate outdoor living
with solid composite
decking

The natural beauty of wood without the maintenance hassle.
Solid nfc decking offers exceptional strength, durability and
weather resistance — easy to stain to your desired colour.

AVAILABLE SIZE	8ft × 6in · 2440 × 150 mm
THICKNESS	25 mm & 30 mm
FINISH	Stainable

View patterns at indowud.com/products/nfc-decking/



07 · NFC FLUTE

nfc flute

Nature's elegance meets modern design

Fluting and louver solutions for green architecture — fostering a harmonious relationship between human habitation and nature, for more sustainable louver and flute panelling.

AVAILABLE SIZE

8ft × 1ft · 2440 × 300 mm

THICKNESS

18 mm & 25 mm

View patterns at indowud.com/products/nfcFlute/



NFC FLUTE · LOUVER

08 · NFC TEXTURED PANEL

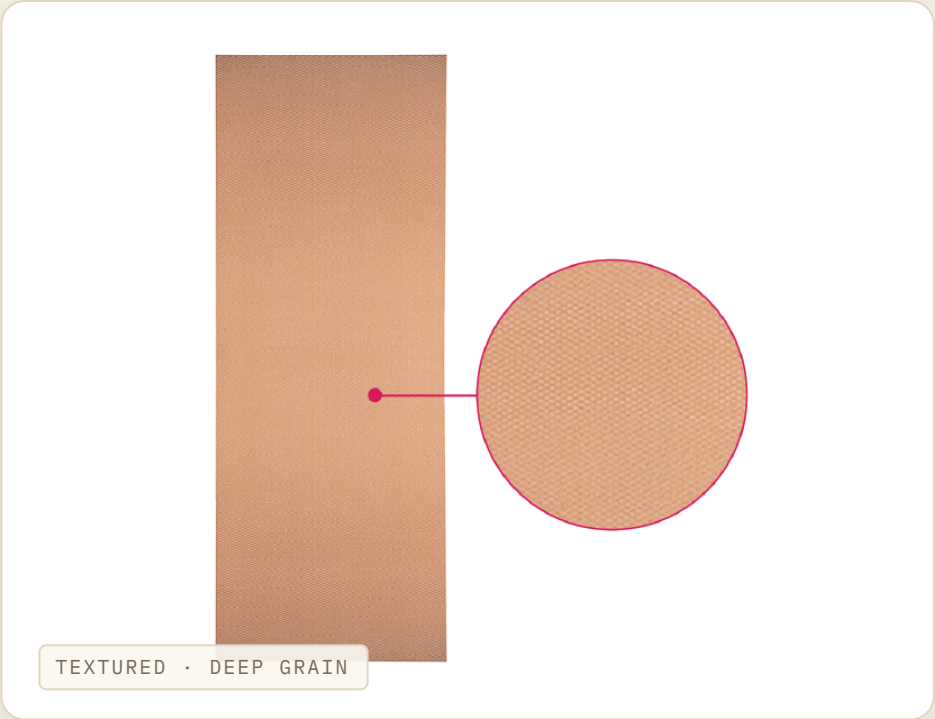
nfc textured panels

The magic of nature in intricate wooden grains

A revolutionary line of natural-fibre composites with stunning wooden-textured designs — the warmth and elegance of wood with the durability and versatility of modern composites.

TEXTURED	8ft × 2ft · 2440 × 600 mm
THICKNESS	8 · 12 · 15 · 18 · 25 mm
DEEP TEXTURE	8ft × 1ft · 13 · 15 · 23 mm

View designs at indowud.com/products/nfctexturedpanels/



09 · NFC TRIM

nfc trim

Natural choice for eco-friendly trims

A sustainable, versatile solution for cut-to-size needs — perfect for adding a touch of elegance to any project. Easily customised to fit your specific requirements.

STANDARD LENGTH	8ft · 2440 mm
THICKNESS	8 · 12 · 15 · 18 · 25 · 30 mm
STANDARD WIDTH	100 · 150 · 200 · 250 mm



10 · NFC FENCE

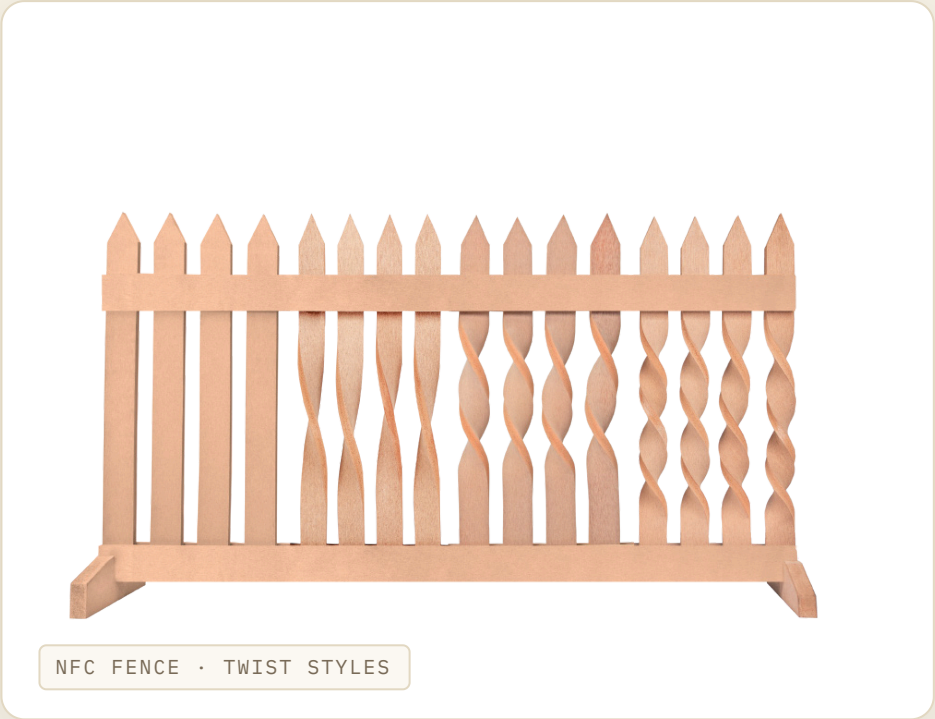
nfc fence

Beauty and sustainability combined

A beautiful, eco-friendly alternative to traditional wood fencing — strong, durable and weather-resistant. Perfect for any outdoor space.

LENGTH	4ft · 1220 mm
WIDTH	50 · 75 · 100 mm
THICKNESS	15 · 18 · 25 mm
STYLE	Standard · single/double/triple twist

View designs at indowud.com/products/nfcfence/



11 · NFC GLU

nfc glu

A strong, water-resistant bond

NFC-GLU bonds almost all surfaces — rough or smooth, porous or non-porous. We strongly recommend using NFC-GLU with Indowud nfc to bond different surfaces together, faster.

- ◊ WATER-RESISTANT
- ✓ UNIVERSAL BOND

BONDS	Rough · smooth · porous · non-porous
BEST WITH	Indowud nfc surfaces

*Subject to availability.



MADE WITH THE AHIMSA DESIGN PHILOSOPHY

Engineered to safeguard the environment.

Made with sustainable agricultural-husk, Indowud nfc is crafted in a world-standard design, research and manufacturing facility in Chennai, India.

CHENNAI, INDIA

WORLD-STANDARD FACILITY

IN-HOUSE R&D

THE PROCESS

Engineering a revolutionary product that surpasses the best global practices.

01



Selection of raw material

We acquire the finest virgin PVC resin from Japan and South Korea, and partner with local farming communities for an uninterrupted supply of natural fibres.

02



Matrix formulation

Polymers and fibres are mixed with minerals, coupling agents, heat stabilisers and additives under monitored temperature and pressure to form a homogeneous bonding matrix.

03



Interface strength

Bonds and density decide each board's interfacial strength — achieved with mathematical accuracy of fibre dispersion for better impact resistance and internal strength.

04



Manufacturing

The compound is extruded under controlled temperature through calibration pads, cooled, then transverse-cut, trimmed and surface-processed.

625 kg/CBM

NEO DENSITY

725 kg/CBM

CREATE DENSITY

825 kg/CBM

BUILD DENSITY

FORTIFIED FOR DURABILITY

Indowud nfc is a high-performance wood.

Fortified for durability — tested, certified, and built to outlast the conditions that defeat ordinary wood.

HANDLING

Design smarter, create faster.

Handling Indowud nfc is as easy as conventional wood — cut, nail, screw, drill, overlay laminate & veneers, print, paint or varnish.

**Screw****Nail****Cut****Paint****Thermoform****Overlay laminate****Print****CNC routing**

SUPPORT & SUGGESTIONS

Changing the world with small, simple steps.

Indowud nfc boards are homogeneous in structure. For larger areas, constructive support / framing is required as per the maximum support distances below (all values in mm).

BOARD THICKNESS	NEO · 625+ KG/CBM	CREATE · 725+ KG/CBM	BUILD · 825+ KG/CBM
6 mm	150	200	250
8 mm	200	250	350
12 mm	250	300	450
15/16 mm	300	350	550
18 mm	350	400	600
20 mm	400	450	650
25 mm	500	550	750

Suggested distances may increase or decrease depending on application.

INSTALLATION NOTES

Get the details right.

- 01 Suggested distances may vary with application.
- 02 Fix a magnetic ball catch at the top and bottom of wardrobe/cabinet shutters. A straightener/stiffener is recommended.
- 03 Use box-type hinges at a span of every 300 mm.
- 04 Fix shutters only after overlaying laminate, veneer, edge-band / lipping or applying epoxy/solvent putty / primer / sealant all around to close micro-pores and avoid bowing.
- 05 Ensure framing on all four sides with stiffeners for sliding doors / shutters.
- 06 Suggested glue: NFC-GLU, PUR, HeatX, Wp1, Probond, Plastilok, Relam. Drying time varies with conditions.
- 07 For ceilings, use 12 mm+ board with a grid support of 300 mm (Create/Build) or 200 mm (Neo).
- 08 For decking, use 25 mm+ with 300 mm support; leave a 4 mm gap between deck panels.
- 09 Do not pour water on raw boards before closing surface and edge micro-pores. Do not use NC putty.
- 10 Leave a 12 mm gap to the wall, 5 mm at ceiling/floor, and 3 mm between boards/trims for breathing and expansion.

THERMOFORMING

Shape it without compromise.

Thermoform Indowud nfc with a thermoforming machine, heat oven, heat gun or infrared heater. High natural-fibre content keeps density and dimensional stability uncompromised post-forming.

Note: the panel may shrink when shaped — do any cutting or work after the board takes its final shape. Prepare a mould for even shaping.

HEATING TEMPERATURE	140 – 160 °C
HEATING TIME	1–2 min / mm of thickness (e.g. 18–36 min for 18 mm)
LOCKING TIME	5–20 min, by thickness & conditions
COOLING TIME	1–2 min / mm of thickness

FINISHING

Finish it any way you like.

Indowud nfc takes print, paint, laminate, veneer and fasteners as readily as fine wood — with holding power wood can't match.



Printing

Any pattern or design can be printed on Indowud nfc with a flatbed digital printer. Ensure the board is dust-free before printing. We recommend a transparent sealant or coat of varnish to protect the printed area from abrasion and dust.



Screwing & nailing

High natural-fibre content enables excellent screw/nail holding. Choose countersunk, wide-head, fully-threaded fasteners. Gluing the edge adds joinery strength. Stainless-steel screws are ideal for outdoor applications.



Laminate, veneer & paint

Overlay laminate or veneer, or paint, polish, stain and varnish directly. Sealing the surface and edges closes micro-pores for a flawless, lasting finish indoors or out.

GREEN RATING

Adds merit to projects with Green Rating.

Indowud nfc is a GreenPro certified product (Oct 2021) contributing recognised credits toward green-building certification.

MAXIMUM POINTS WITH INDOWUD · MODULE COMPLIANCE		CREDIT
Wood-based material with FSC certification and/or rapidly renewable		3 Points
Local materials		3 Points
Use of certified green building materials		3 Points
Composite wood with no urea formaldehyde		1 Point
Materials with recycled content		1 Point

Contact us for more details · indowud.com/sustainability-green-rating

COMPARATIVE STUDY OF NFC WITH OTHERS

Quality the world and time can test.

Available 8'x4' in 6–25 mm and 7'x3' in 28/30 mm; custom sizes available. Density may vary by 5%.

PROPERTY	NFC	PVC / WPC FOAM	PLYWOOD	MDF
Density (kg/CBM)	650–800	400–600	650–750	600–700
Raw materials	Natural fibres + thermoplastics	PVC & fillers	Medium/soft wood, urea, formaldehyde	Medium/soft wood, urea/phenol
Termite proof	Yes	Yes	No	No
Water proof	Yes	Yes	No	No
Flame resistant	Yes	Not always	No	No
Screw holding	Above par	Above par	Below par	Below par
Conventional tools	Yes	Yes	Yes	Yes
Overlay laminate/veneer	Yes	Not always	Yes	Yes
Indoor & outdoor	Both	Preferably indoor	Only indoor	Indoor
Weather & ageing resistant	Above par	For some time	No	No
Eco-friendly	Yes	Hazardous	No	No

NFC BOARDS WITH HIGH FIBRE CONTENT

Tested, independently.

Tests carried out on Indowud nfc samples by the National Test House, CIPET, IPIRTI, Spectro Labs and SGS Labs.

TEST	UNIT	METHOD	RESULT
Density	Kg/CBM	ASTM D792	800.349
Water absorption · 2 hrs	%	IS 2380	0.02
Water absorption · 24 hrs	%	IS 2380	0.12
Thickness swelling · 2 hrs	%	IS 2380	0.06
Modulus of rupture · avg	N/mm ²	IS 2380	14.5
Modulus of rupture · min	N/mm ²	IS 2380	14.4
Modulus of elasticity · avg	N/mm ²	IS 2380	1327
Modulus of elasticity · min	N/mm ²	IS 2380	1290
Screw withdrawal · face	N	IS 2380	2252
Screw withdrawal · edge	N	IS 2380	1409

TEST	UNIT	METHOD	RESULT
Tensile strength	MPa	ASTM D638	7.6
Compression strength	MPa	ASTM D695	40.2
Elongation @ break	%	ASTM D638	2.0
Charpy impact strength	KJ/m ²	ASTM D6110	6.25
Heat deflection @ 0.45 MPa	°C	ASTM D648	64.45
Softening temp @ 1 kg	°C	ASTM D1525	72.5
VOC emissions	Mg/Kg	EPA 5035A	Below detectable
Termite resistance	—	IS 4833	No attack
Fungal resistance	—	IS 4873	No attack
Borer resistance	—	IS 4873	No attack

ACOUSTICS

Quiet, by construction.

Sound absorption per IS 10420:1982 and transmission loss per ISO 10534-2:1998.

FREQUENCY (HZ)	125	250	500	1000	1500	2000	2500	3000	3500	4000
Sound absorption co-eff.	0.02	0.03	0.04	0.06	0.06	0.05	0.05	0.06	0.08	0.10
Transmission loss (dB)	26.19	22.53	32.85	40.2	44.5	47.33	45.61	47.83	46.6	47.56

40.12

SOUND TRANSMISSION CLASS (AVG)

0.10

PEAK ABSORPTION @ 4000 HZ

FIRE TEST

Flame-retardant & smoke-suppressant.

UIC 564-2 (Appendix 11) governs fire protection in railway vehicles. UL94 V0 means burning stops within 10 seconds. ASTM E84 assesses surface burning — flame spread and smoke developed indices.

TEST	METHOD	LIMIT	RESULT
Fire resistance	Appendix 11 of UIC 564.2	Class A	Class 1 or A
Flammability	UL94	V0 rating	V0 Rating
Flame spread index	ASTM E84:2020	0 – 25	6
Smoke developed index	ASTM E84:2020	≤ 450	100

Tested to international standards · more at indowud.com/Fire-test/

TOGETHER

Let's build a world where beauty and responsibility go hand in hand.

REGD. OFFICE

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